Laboratory Manual SCOPE Department 2025-2026

Contents

Experiment 1: Comprehensive Water Analysis 2

Experiment 2: Removal of heavy metals from waste water using nanoparticles prepared from bio-waste 7

Experiment 3: Building Flexible Dye Sensitized Solar Cells as Renewable Resource: Natural Dye and TiO_2 from Waste Products 12

Experiment 4: Construction of galvanic cells using waste metal and evaluation of Thermodynamic functions (enthalpy, entropy, and free energy) from EMF measurements 14

Experiment 5. Determination of reaction rate, order and molecularity of ethylacetate hydrolysis 20

Experiment 6. Magnesium Salt Water Activated Battery-Effect of Salt Concentration on Voltage generation 24

Experiment 7. Size dependent color variation of Cu_2O nanoparticles by a spectrophotometer 27

Experiment 8. Synthesis of Al-doped ZnO Nanoparticles via Sol-Gel Method 31

Experiment 9. Kinetic Study of Rhodamine B Degradation by •OH Radicals (H_2O_2/UV System) 35

Experiment 10. Determination of State of Charge (SoC) 40

Experiment 1: Comprehensive Water Analysis

Aim:

Collect a locally available water sample (suspected for contamination) and analyze its hardness, chloride content, Total Dissolved Solids (TDS), pH, and sulfate content in it. After analyzing compare the water quality as specified by Central Pollution Control Board, Ministry of Environment, Forest and Climate Change, Government of India (https://cpcb.nic.in/wqstandards/)

Principles:

a) Hardness Estimation: Ehtylenediaminetetraacetic acid (EDTA) forms complexes with a large number of cations including Ca2+ and Mg2+ depending upon pH of solution. Hence, it is possible to determine the total hardness of water using EDTA solution. EDTA in the form of its sodium salt (H2Y2-) is commonly used in complexometric titration for estimation of metal ion because pure EDTA (H4Y) is sparingly soluble in water. EDTA has six binding sites (the four carboxylate groups and the two amino groups) providing six pairs of electrons. The resulting metal-ligand complex, in which EDTA forms a cage-like structure around the metal ion, is very stable at specific pH. All metal-EDTA complexes have a 1:1 stoichiometry. The H2Y2- form complexes with metal ions as follows.

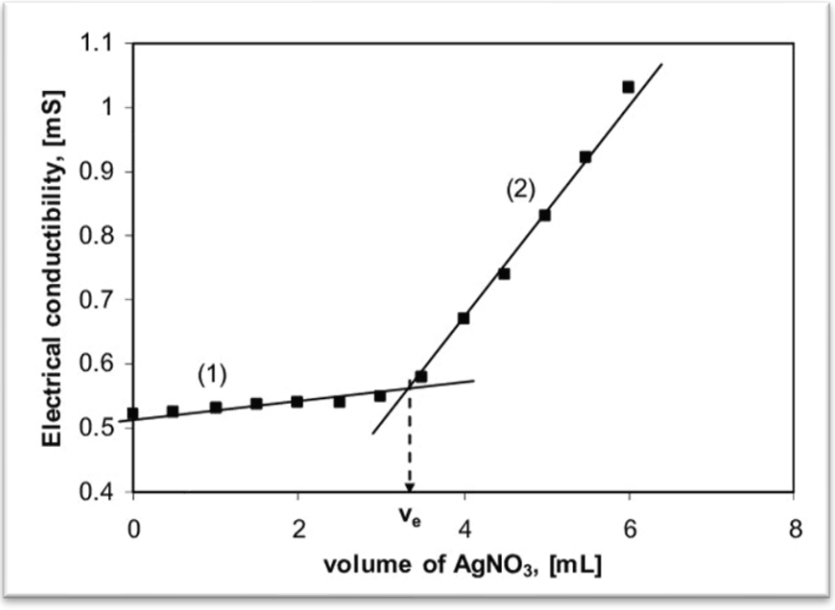
$$M + H_2Y^{2-} \rightarrow MY^{2-} + 2H^+ \qquad (1)$$

Where, M is Ca2+ and Mg2+ present in water. Reaction (1) can be carried out quantitatively at pH 10 using Eriochrome Black T (EBT) as indicator. EBT forms a wine-red complex with M2+ ions which is relatively less stable than the M2+-EDTA complex. On titration, EDTA first reacts with free M2+ ions and then with the metal-EBT indicator complex. The latter gives a colour change from wine-red to steel blue at the equivalence point. *(reference present Manual)*

b) Chloride Estimation: the methods considered for chloride analysis in this study are based on the well-known reaction between chloride ions and AgNO3, when a white precipitate of AgCl is formed, according with the following equation:

$$Cl^- + AgNO_3 = AgCl + NO_3^- \quad (2)$$

The obtained precipitate has a low solubility in water. In addition, because only halide and sulfide ions precipitate in presence of $AgNO_3$, such methods have a good selectivity in quantitative analysis of chloride ions. The representative titration graph is given below.



c) Total Dissolved Solids (TDS) estimation: The use of electrical conductivity (EC) is based on the observed linear relationship between specific conductance and ionic strength for fluids with conductance ranging from 10–10,000 µS/cm. A common expression is:

$$TDS\ (mg/L) = k_e \times EC(\mu S/cm) \quad (3)$$

where k_e is a constant of proportionality. The TDS-EC relationship is affected by the ionic composition of the water and the concentration of dissolved species. For most natural waters, the k_e value varies from 0.55 to 0.85, and the average of these two extremes (0.7) is widely used. (Reference: https://pure.psu.edu/en/publications/relationship-between-total-dissolved-solids-and-electrical-conduc or https://doi.org/10.2166/wst.2018.092)

d) pH Estimation: The term "pH" refers to the measurement of hydrogen ion activity in the solution. Since the direct measurement of the pH is very difficult, specific electrodes are needed for quick and accurate pH determination. pH is measured on a scale of 0 to 14, with lower values indicating high H+ (more acidic) and higher values indicating low H+ ion activity (less acidic). A pH of 7 is considered as neutral. Every whole unit in pH represents a ten-fold increase in or decrease in hydrogen ion concentration. Most natural waters possess the pH values ranging from 5.0 to 8.5. Rain water have a pH value of 5.4 to 6.0 which then reacts with the soils and minerals causing the reduction in H+ ion concentration and thus the water may become alkaline with a pH 0f 8.0-8.5. More acid water (pH<5) and more alkaline (pH >9) and other immediate changes in the hydrogen ion concentration (pH) suggest that the quality of the water is adversely affected due to the introduction of some toxic contaminants in water bodies. pH is measured using pH meter, which comprises a detecting unit consisting of a glass electrode, reference electrode, usually a calomel electrode connected by KCl Bridge to the pH sensitive glass electrode and an indicating unit which indicates the pH corresponding to the electromotive force is then detected. Before measurement, pH meter should be calibrated by using at least two buffers. (Reference: https://nitsri.ac.in/Department/Civil%20Engineering/CWE-301_WATER_QUALITY_AND_ENVIRONMENT_pH_Meters.pdf)

e) Sulfate Estimation: Sulfate ion is converted to a barium sulfate suspension under controlled conditions. The resulting turbidity is determined by a nephelometer, filter photometer, or spectrophotometer and compared with a curve prepared from standard sulfate solution. (Reference: https://www.epa.gov/sites/default/files/2015-12/documents/9038.pdf)

$$BaCl_2 + SO_4^{2-} = BaSO_4 \downarrow + 2Cl^- \quad (4)$$

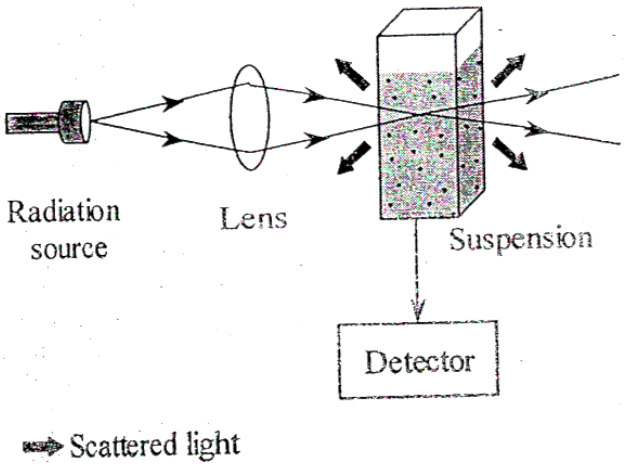


Figure : Shcematic Diagram Representing Light Scattering in Nephelometry

Requirements:

a) Hardness estimation: Reagents and solutions: Standard hard water (1mg/mL of $CaCO_3$ equivalents), 0.01 N EDTA solution, EBT indicator, hard water sample, NH3-NH4Cl buffer solution.

Apparatus: Burette, pipette, conical flask, standard flask, and burette stand.

b) Chloride Estimation: Reagents and solutions: Standard $AgNO_3$, and NaCl solutions.

Apparatus: Burette, pipette, beaker, standard flask, burette stand, Pt electrode, a digital conductometer and magnetic stir bar

c) Total Dissolved Solids (TDS) estimation: Reagents and solutions: None. Apparatus: Commercial TDS meter, Pt electrode, and a digital conductometer.

d) pH Estimation: Reagents and solutions: pH buffer solution for calibrating pH, and pH paper. Apparatus: Digital pH Meter along with electrodes. Commercial TDS meter, Pt electrode, and a digital conductometer.

e) Sulfate Estimation: Reagents and solutions: conc. HCl, Isopropanol, NaCl, Propanol, glycerol, Na_2SO_4, and $BaCl_2$.

Apparatus: Magnetic stirrer, Nephelometer, Stopwatch, Measuring spoon, 250 mL conical flask, and 1L volumetric flask.

Procedures:

a) Hardness estimation:

Pipette out 20 mL of the given sample of water into a clean conical flask. Add one test tube full of ammonia buffer (NH_4OH – NH_4Cl) solution and three drops of Eriochrome Black– T (EBT) indicator. Titrate this mixture against standardized EDTA solution taken in the burette. The end point is the change of color from wine red to steel blue. Repeat the titration for concordant titer value. Let 'V2' be the volume of EDTA consumed.

S. No.	Volume of sample hard water (ml)	Burette reading (ml)		Volume of EDTA (V_2, ml)
		Initial	Final	
1				
2				
3				
Concordant titre value (ml)				

Calculations:

If EDTA is standardized as 1 mL of EDTA requires = w mg $CaCO_3$ for complexation

From the above titration 20 mL of sample water consumes = V2 mL of EDTA = V2 x w mg of $CaCO_3$ eq.

\1000 mL of hard water sample consumes = V2 x w × 1000/20 = V2 x w × 50 ppm

\Hardness of the water sample = "V2 x w × 50" ppm

b) Chloride Estimation:

Make up the unknown water solution given in a 100 mL standard flask up to the mark using distilled water. Pipette out 20 mL of this solution into a 100 mL beaker and add 10 mL distilled water to it. Dip the conductivity cell and fill the burette with ~0.01 N $AgNO_3$ solution (from Bottle B). Record the conductivity of the water sample without adding any $AgNO_3$ solution from the burette (0th reading). Add 1 mL $AgNO_3$ solution of known concentration into the beaker with continuous stirring by magnetic stirrer. Continue the addition of $AgNO_3$ solution (1 mL each time) and note the conductance after each addition. Continue the titration beyond the equivalence point for about 5 mL. The conductance will either decrease slightly or remain constant until complete precipitation of AgCl, and then starts increasing on continuing the addition of $AgNO_3$. A graph is now drawn by plotting conductance vs volume of $AgNO_3$ added. Intersection point from the plot gives the volume of $AgNO_3$ required for precipitating the chloride present in the water sample.

Estimation of unknown chloride:

(N x V) of water sample = (N x V) of $AgNO_3$ solution

N of irrigation water sample = N of $AgNO_3$ x Volume measured from Plot (Ve)

20 mL= N of water sample

Amount of chloride present in 1L = Normality of irrigation water sample x Eq. wt. of Cl^- (35.453)

Amount of chloride present in given sample solution = $\frac{Strength\ of\ water\ sample\ x\ 35.453\ x100}{1000}$

= grams in 100 mL

Result: Amount of chloride in given water sample = $Strength\ of\ water\ sample\ x\ 35.453\ x1000\ ppm$

c) Total Dissolved Solids (TDS) estimation: Recorded the conductivity of the water sample without adding any $AgNO_3$ solution from the burette (0th reading). The recorded conductivity when multiplied by 0.7 (k_e) as per the equation

$TDS\ (mg/L) = k_e \times EC(\mu S/cm)$

TDS is estimated in terms of ppm.

d) pH Estimation: The pH electrode to be dipped in a beaker of sample water and pH will be displayed and to be recorded. Avoid any kind of contamination and use a clean and dry beaker before pouring the sample water.

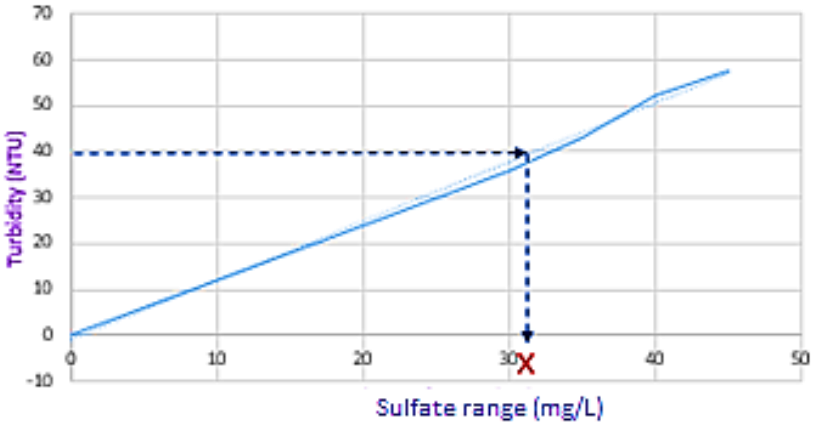
e) Sulfate Estimation:

i.) Preparation of conditioning reagent: Slowly add 30 mL concentrated HCl to 300 mL of DI water, 100 mL 95% ethanol or isopropanol, and 75 g NaCl in solution in a container. Add 50 mL glycerol and mix.

ii.) Formation of barium sulfate turbidity: Place a 100-mL sample, or a suitable portion diluted to 100 mL, into a 250-mL Erlenmeyer flask (conical flask). Add exactly 5.0 mL conditioning reagent and mix in the stirring apparatus. While the solution is being stirred, add a measured spoonful of $BaCl_2$ crystals and begin timing immediately using stop watch. Stir exactly 1.0 min at constant speed.

iii.) Measurement of barium sulfate turbidity: Immediately after the stirring period has ended, pour solution into absorbance cell. Measure turbidity at 30-sec intervals for 4 min. Record the maximum reading obtained in the 4-min period.

iv.) Preparation of calibration curve: Dissolve 147.9 mg anhydrous Na_2SO_4 in DI water in a 1-liter volumetric flask and dilute to the mark for standard sulfate solution. Space standards at 5-mg/L increments in the 0-40 mg/L sulfate range. Prepare a standard curve by plotting turbidity (NTU units) vs the sulfate range. From the calibration plot, measure the sulfate concentration in the given unknown sample.



Report of Water Sample Analysed			
Sample Collected from			
Date of Collection			
Parameters	**Recorded Values**	**Recommended Values as per BIS (Bureau of Indian Standards)**	**Deviation from the standard (%)**
Hardness (ppm)			
Chloride content (ppm)			
TDS (ppm)			
Sulfate content (ppm)			
pH			

Experiment 2: Removal of heavy metals from waste water using nanoparticles prepared from bio-waste

Aim:

To synthesize and characterize hydroxyapatite nanoparticles from eggshell bio-waste by chemical precipitation and study its application for Pb (II) removal from aqueous wastewater.

Principles:

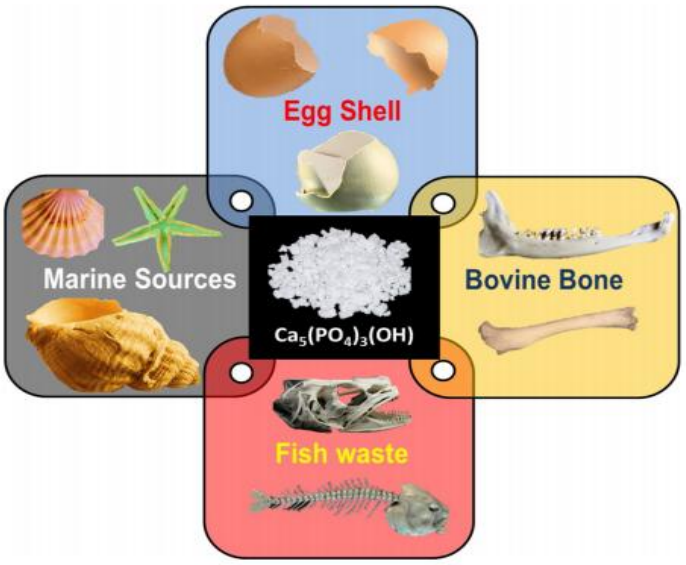
Hydroxyapatite (HAP) nanoparticles find application in various fields such as bio-ceramics, medicine, the pharmaceutical industry, and environmental pollution control. Hence, it is of great importance to synthesize HAP nanoparticles. HAP nanoparticles from eggshell bio-waste were prepared by chemical precipitation method and characterized by powder X-ray diffraction (powder XRD). Hydroxyapatite (HAP), is chemically represented as $Ca_{10}(PO4)_6(OH)_2$. HAP is a bioactive ceramic that has gained wide attention in recent years owing to its potential applications in several fields, such as bio-ceramics, medicine, the pharmaceutical industry, and environmental pollution control [2]. The Ca/P ratio of stoichiometric hydroxyapatite is usually 1.67, whereas the Ca/P ratio of non-stoichiometric calcium deficient hydroxyapatite is below 1.67. Compared to synthetic HAP, natural HAP is non-stoichiometric because it contains trace amounts of Na^+, Zn^{2+}, Mg^{2+}, K^+, Sr^{2+}, Ba^{2+}, F^-, and CO_3^{2-}. Calcium-deficient non-stoichiometric HAPs derived from natural sources have been shown to be better adsorbents for the purification of aqueous solutions containing metal cations, anions, and organic pollutants because of the presence of exchangeable cations, hydroxyl ions, and phosphate chelating groups [3]. HAP can be synthesized from both chemical and natural sources. Currently, studies have focused on the use of bio-waste sources, such as coral, egg shells, fish bones, and seashells, for the preparation of HAP. Among the above sources, eggshell waste is considered a potential material because of its large availability and rich calcium content (> 95wt% $CaCO_3$) [4].

Requirements:

- Cleaned the eggshell
- Concentrated HNO_3
- NaOH
- $Ca(NO_3)_2$
- $(NH4)_2HPO_4$

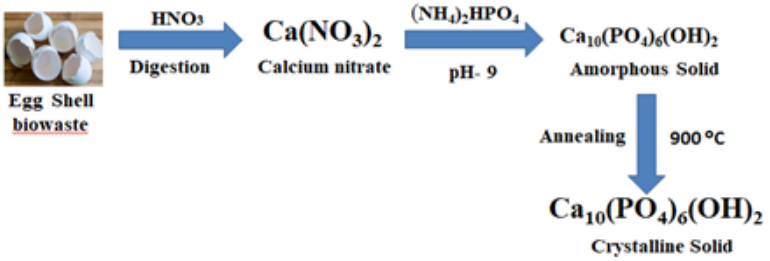
Procedures:

Natural bio-waste precursors for HAP synthesis- HAP nanoparticle synthesis from eggshell bio-waste:



Hydroxyapatite nanoparticles are prepared from eggshell bio-waste by simple chemical precipitation method. First, clean the eggshell biowaste with tap water several times followed by distilled water to remove dirt and organic residues from the eggshell and then dry it in hot air oven at 100°C. Crush the cleaned and dried eggshell bio-waste using agate mortar to get a fine powder of eggshell. Weigh around 1g of powdered eggshell bio-waste and transfer into a 100 ml beaker and then add 5 mL of concentrated HNO_3 slowly. Stir the mixture continuously until the eggshell powder dissolves completely in nitric acid. To this, add 15 ml of deionized water followed by 20 ml of 4N NaOH solution to get yellow coloured colloidal solution. The resultant yellow turbid solution is passed through filter paper to get clear aqueous solution of $Ca(NO_3)_2$. A white milky precipitate of amorphous hydroxyapatite are obtained by adding slowly 0.1 M of $(NH_4)_2HPO_4$ to $Ca(NO_3)_2$ solution (pH of the solution maintained at 9). The precipitate is then filtered through Whatman filter paper and washed several times with deionized water to remove any excess untreated precursors and byproducts (NH_4^+ and NO_3^- ions). Finally, the precipitate is allowed to dry for 1 h at 100 °C in a hot air oven followed by annealing in an air atmosphere at 900 °C for 3 h using an electrical furnace to get crystalline hydroxyapatite [$Ca_{10}(PO4)_6(OH)_2$].

Scheme:



Characterization:

The powder XRD pattern of the collected powder sample was recorded on a Rigaku Smartlab 3 KW X-ray diffractometer with Cu Kα radiation source (λ=1.54060 Å)) in the scan range 10-80° (2θ). The standard XRD pattern of HAP is given in Fig. 1 for reference.

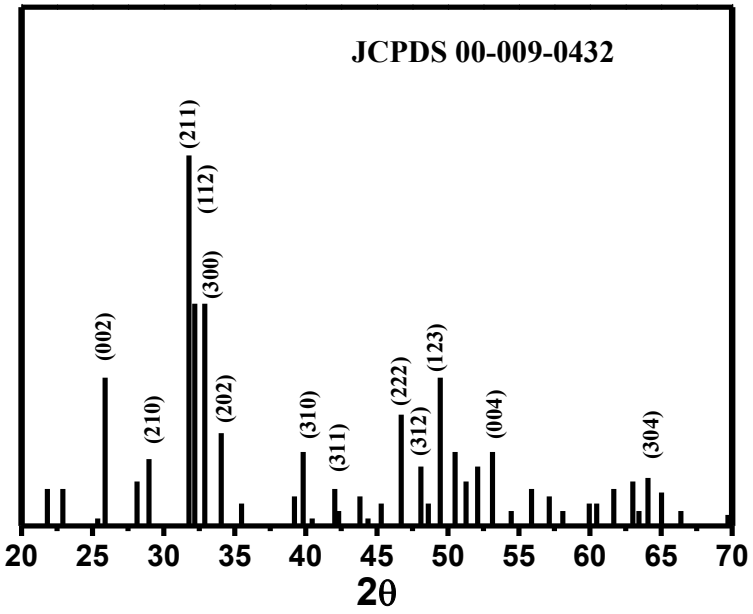
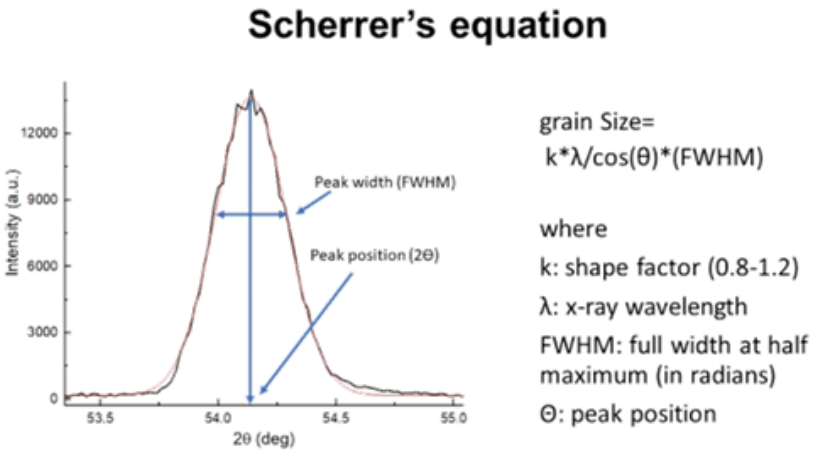


Fig.1. XRD pattern of HAP nanoparticles

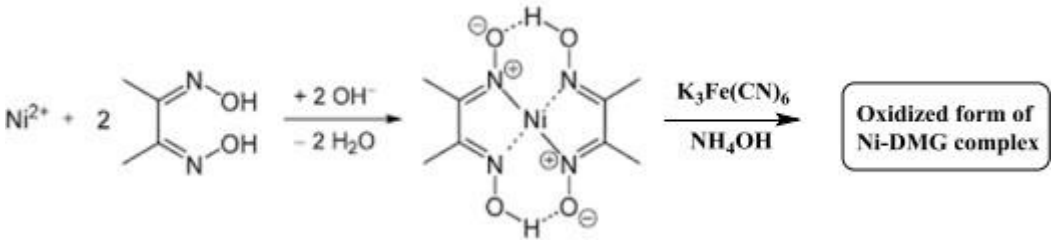
Analysis:
The powder XRD peak positions for the prepared HAP nanoparticles by chemical precipitation method are identified based on standard **JCPDS file #__09-0432.**

Particle size calculation using Scherrer's equation:

Scherrer equation
Grain size = k*λ / cos(θ)*(FWHM)
k = 0.9, λ = 1.0506 Å, and "θ" and FWHM will be obtained from powder XRD data.



Principle: Dimethylglyoxime (DMG) reacts with nickel ions and forms a pink-colored Ni(DMG)2 complex in alkaline medium. It gets oxidized by potassium ferricyanide (K3[Fe(CN)6]) in alkaline medium to form a brown-red, water soluble oxidized Ni(DMG)2 complex as shown in the Scheme 2. Absorption spectrum of the oxidized complex shows absorption maxima at a wavelength of 440 nm (Fig.4).



Scheme 2: Formation of brown-red, water soluble oxidized $Ni(DMG)_2$ complex.

Requirements:

Reagents and solutions: $NiSO_4$ (1000 ppm), NaOH (1 N) solution, DMG (2 % in 2 N NaOH), $K_3[Fe(CN)_6]$ (1.5 %).

Apparatus: Colorimetry, Volumetric flasks, Beakers.

Procedure: Take 6 standard 50 mL volumetric flasks (including the unknown). To every flask add 0.5 mL of DMG solution followed by 0.5 mL of $K_3[Fe(CN)_6]$ solution using a burette. Further, add 0.5, 1, 1.5, 2 and 2.5 mL of the working Ni stock solution (1000 ppm) from a burette to get 10, 20, 30, 40 and 50 ppm of steel containing nickel (II) solutions. All the flasks are shaken well and made up to 50 mL mark with 1N NaOH solution. Allow the flasks at least 15 minutes after adding all the reagents for the complete complex formation. Then the absorbance of the formed brown-red solution is measured at 440 nm against distilled water (blank). Record these absorbance readings in Table 1. Draw a calibration graph taking concentration of Ni^{2+} (in ppm) as X-axis and absorbance readings as Y-axis. A straight line that passes through the origin (Fig.5) is an indication that the measured data obeys Beer's Law. From the calibration plot, measure the concentration of nickel in the given unknown sample.

(ii) Removal of Ni^{2+} ions from wastewater using HAP nanoparticles

HAP can be employed to effectively remove Nickel (Ni^{2+}) ions from wastewater in batch adsorption studies. First, HAP (0.1 g) was dispersed in 30 ml of deionized water and added to 10 mL of wastewater containing Ni^{2+} ions. The mixture was shaken at 180 rpm for 1 h at room temperature in an orbital shaker. Subsequently, the HAP was filtered using Whatman filter paper, and the residual Ni^{2+} content in the filtrate was estimated using a dimethylglyoxime (DMG) colorimetric reagent at a wavelength of 440 nm detailed above. Finally, the percentage of Ni^{2+} ions adsorbed on HAP nanoparticles was calculated based on the initial and final concentrations of Ni^{2+} ions in the wastewater using the following equation: % $Removal$ = $(C_0 - C_e) \times 100 C_0$

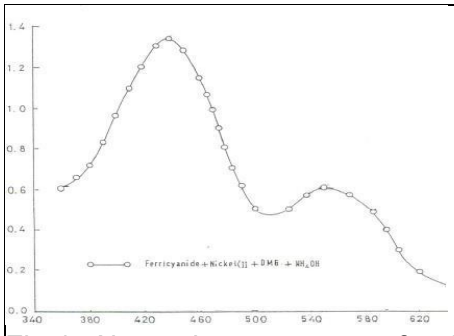


Fig.4: Absorption spectrum of oxidized Ni(II)-DMG complex showing λmax at 440 nm

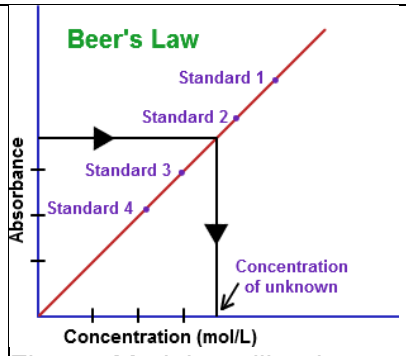


Fig.5: Model calibration curve for Ni(II) determination

Table 1: Experimental Data:

S. No.	Concentration (ppm) (X-axis)	Absorbance (Y-axis)
1.		
2.		

3.		
4.		
5.		
6.	Unknown	

Results:

(i) The powder XRD peak positions for the prepared HAP nanoparticles are

_____

(ii) Crystallite size of HAP is ___

(iii) The percentage removal of Ni^{2+} ions from wastewater ……

Experiment 3: Building Flexible Dye Sensitized Solar Cells as Renewable Resource: Natural Dye and TiO_2 from Waste Products

Aim:
To harvest solar energy by using TiO_2 and readily available natural dye from pomegranate, beet root, carrot *etc.* along with flexible ITO coated and generates electrical energy.

Principle:
Many food and personal care products usually contents substantial amount of TiO_2 in them. Once they become waste products, TiO_2 in them can be extracted and utilized for electron acceptor as photo anode for dye sensitized solar cell (DSSC). In DSSCs light is absorbed by a monolayer of a dye located at the junction between an electron and hole transporting phases where the electron transport material is a wide band gap semiconductor material typically TiO_2 (anatase), while the hole transport system is a redox shuttle. On light-irradiation the dye injects an electron and a hole into the n- and p-type materials, respectively, generating free charge carriers, which then travel through the nanostructures to be collected as current and flow to the external circuits. During the last two decades research efforts have been made on the development of new sensitizers, photoanode engineering, redox shuttle and catalyst to fabricate efficient devices. The DSSC consists of a mechanical support coated with a transparent conductor, on which TiO_2 nanocrystals are deposited to form a mesoporous network on which a sensitizer is anchored, and in combination with platinized cathode separated by a thin layer of redox shuttle.

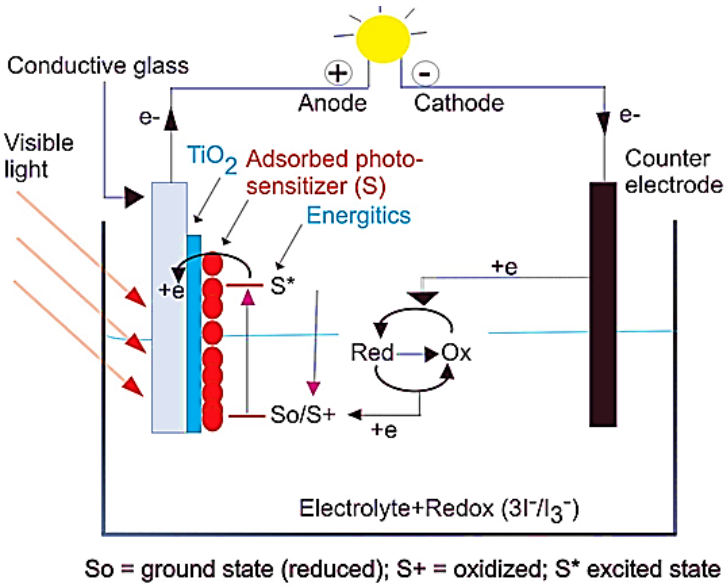


Fig.1: Operating principles and energy level diagram of a dye-sensitized solar cell with a polymer supported cathode.

Requirements:
Reagents and solutions: ITO coated PET, natural dye, TiO2 powder, conducting charcoal. KI, I_2 solution, EtOH, and polymer adhesive, transparent PET sheet, selected food and personal care product having more TiO_2 content

Apparatus: Multimeter, Hotplate. 100 mL beakers, glass rod, 5 mm multicolor LEDs, 70 Points Mini Breadboard, Solderless Mini Breadboard for Prototyping, solid core wire for breadboard, and Alligator (Crocodile) Clip

Procedure:
Commercially available ITO coated PET sheet has to be coated with TiO_2 ethanol slurry. Then the coated film will be dried and soaked with natural dye (pomegranate juice). The soaked device will be dried or backed at 70 °C for 5 minutes. The conducting charcoal or Carbon slurry will be coated on transparent PET sheet or

paper, and dried before soaking it with $KI+I_2$ solution. The conducting carbon film and the TiO_2 and dye soaked films were clamped and connectors were added from individual films. The transparent side of ITO coated film is illuminated under sunlight and the voltage drop will be measured through multimeter. The galvanic cell prepared to be connected to solderless breadboard for prototype circuit where 5mm multicolor LEDs glow.

Results:

Number of cells made	Number of LEDs glow	Duration of glow

Experiment 4: Construction of galvanic cells using waste metal and evaluation of Thermodynamic functions (enthalpy, entropy, and free energy) from EMF measurements

Experiment	**Thermodynamic functions from EMF measurements: Zinc-Copper system**
Aim	To construct and to measure electromotive force (EMF) of a Daniell cell at varied concentrations of metal ions and temperatures To determine the thermodynamic functions (**ΔG, ΔH, and ΔS)**
Problem definition	Construction of a Daniell Cell, measurement of EMF and determination of **ΔG, ΔH, and ΔS**
Solution	The measurement of EMF will be carried out by a **Potentiometer** and followed by the calculation of different thermodynamic functions through the EMF values
Student learning outcomes	Students will be able: • To understand the construction of a Daniell cell (Zn-Cu system) • To observe the influence of metal-ion concentration and temperature on the EMF on Zn-Cu system • To calculate the thermodynamic functions **(ΔG, ΔH,** and **ΔS**)

Principle: The electromotive force (EMF) of an electrochemical cell is measured by means of a potentiometer. An electrochemical cell (E_{cell}) is considered as a combination of two individual single cell-electrodes. The potential difference between the two single electrode potentials is a measure of EMF of the cell ($E_{cell;}$ Cathode-Anode/V). In order to measure the potential difference between electrodes in contact with electrolyte containing the same cation, it is necessary to have another electrode in contact with electrolyte of same cation, both the half-cells connected through a salt bridge. Saturated calomel electrode (SCE; $E_{Calomel,}$ E^o = 0.244 V) whose potential is known, is used as **a reference electrode** (Anode; +ve terminal, black-colored wire) and it is coupled with the metal electrode (Cathode, -ve terminal; red-colored wire) for which the potential is to be determined. The substance that loses electrons is said to be oxidized, while the one gaining electrons is reduced. Thus if a piece of zinc metal were immersed into a solution containing copper (II) ions, zinc would be oxidized by copper (II) ions. Zinc loses electrons and is oxidized, and the copper (II) ions gain electrons and are reduced. We can conveniently express these processes together as a Daniell cell by the following reactions (Figure 1):

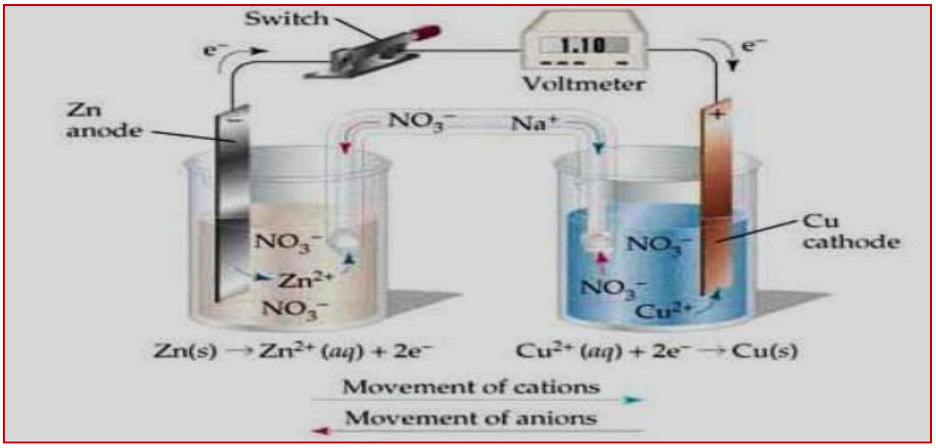


Figure 1: Functioning of a Daniell cell (Zn-Cu) using a salt bridge

In the Daniell cell, copper and zinc electrodes are immersed in the equimolar solution of copper(II) sulfate and zinc(II) sulfate respectively. At the anode, zinc is oxidized as per the following half-reaction: $Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e^-$ (E^o = -0.760 V)

At the cathode, copper is reduced as per the following reaction: $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$ (E^o = +0.340 V)

The overall reaction is: **$Zn_{(s)} + Cu^{2+}_{(aq)} \rightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$ E_{cell} = 0.340 – (-0.760) = 1.10 V**

Importance of Thermodynamic functions:

∆G - Gibbs free energy determines whether a given chemical change is thermodynamically possible or not. The change in Gibbs free energy (**∆G**) is the symbol for spontaneity, and there are two factors which can affect it: enthalpy and entropy. Enthalpy - the heat content of a system at constant pressure. A negative ∆G means that the reactants, or initial state, have more free energy than the products, or final state.

∆H - Measuring the change in enthalpy allows us to determine whether a reaction was endothermic (absorbed heat, positive change in enthalpy) or exothermic (released heat, a negative change in enthalpy.) It is used to calculate the heat of reaction of a chemical process.

∆S - Entropy is the measure of a system's thermal energy per unit temperature that is unavailable for doing useful work. Entropy has been regarded as a measure of disorder or randomness of a system. Thus when a system goes from a more orderly to less orderly state, there is an increase in its randomness and hence entropy of the system increases.

Part-A: Determination of single electrode potentials of M/M^{n+} system ($E^o_{M/Mn+}$) at two different concentrations

E_{cell}: Hg / Hg_2Cl_2 (s), saturated KCl (Anode) ‖ (N/10) electrolyte of the Metal / Metal (Cathode)

From the EMF of the cell involving saturated calomel electrode and metal electrode dipped in its solution of 0.1 M electrolyte, electrode potential of the metal electrode is readily calculated using the standard potential of calomel electrode as E_{cell} (measured from potentiometer) = E_{M/M^+} (Cathode, V) – $E_{calomel}$ (0.244V, Anode) or $E_{M/M^+} = E_{cell} + E_{calomel}$. E_{cell} is the total EMF, i.e., net voltage of the cell measured using a potentiometer. Then, the electrode potential of the metal electrode (E^o_{M/M^+}/ V) can be calculated after incorporating the observed E_{M/M^+} and $a_{M^{n+}}$ parameters in Nernst equation as given below:

Requirements: *Reagents and solutions:* Copper sulphate stock solution (0.1 M), Zinc sulphate stock solution (0.1 M), KCl salt; *Apparatus:* Digital potentiometer, copper electrode, zinc electrode, calomel electrode, 100 mL beaker, burette, standard flask.

$$E_{M/M^+} = E^\circ_{M/M^+} + \frac{RT}{nF} \ln a_{M^{n+}}$$

$$E^\circ_{M/M^+} = E_{M/M^+} - \frac{RT}{nF} \ln a_{M^{n+}}$$

$$E^\circ_{M/M^+} = E_{M/M^+} - \frac{0.0595}{n} \log a_{M^{n+}}$$

Procedure: Calibrate the digital potentiometer with the help of an inbuilt battery support to display 1.018 V. The metal electrode is sensitized by dipping in a small quantity of 1:1 nitric acid containing a small quantity of sodium nitrite until effervescence occurs. Then the electrode is washed well with distilled water. 50 mL of the given concentration of the electrolyte solution is taken in a beaker and its corresponding metal electrode is introduced. This is connected with the saturated calomel electrode (half-cell) by means of a salt bridge. The metal electrode is connected to the positive terminal of and the calomel electrode is connected to the negative terminal of the potentiometer. The EMF of the cell (E_{cell}) is measured and noted in Table 1. The standard electrode potential [$E^\circ_{M/M2+}$] is computed using Nernst equation (Eq. 1).

Table 1: EMF measured for various concentrations of M/M^{n+} system

Electrode/ Electrolyte	Concentration of Electrolyte (N)	E_{cell} (V) (EMF of the cell)	$E_{M/M+} = E_{cell} + E_{calomel}$	$E^\circ_{M/M+}$/V (from Nernst Eqn-1)
Zn/Zn^{2+}	0.05 M			
	0.1 M			
Cu/Cu^{2+}	0.05 M			
	0.1 M			

Table 2: Individual activity coefficients of Cu^{2+} and Zn^{2+} in water at 25 °C

Metal ion system (Cu^{2+}/Zn^{2+})	0.001	0.002	0.005	0.01	0.02	0.05	0.1	0.2
Activity coefficient (γ_c)	0.905	0.870	0.809	0.749	0.675	0.570	0.485	0.405

where E° is the standard electrode potential of the metal electrode; $a_{M^{n+}}$ is the activity of metal ions in the solution ($a_{M^{n+}} = \gamma_c[C]$); γ_c is the activity coefficient (Table 2) and C is the concentration of the electrolyte solution. Solution Temperature (T) = 27 °C

The potential of saturated calomel electrode (SCE)= 0.244 ± 0.0007 (25°C)

$$\mathbf{E^\circ_{M/M^+} = E_{M/M^+} - \frac{0.0595}{n} \log [\gamma_c \times C]} \quad \text{------ Eq.1}$$

The EMFs can be calculated by the Nernst equation (Eq.1).

Part-B: Construction of Daniell cell and measurement of its voltage with three different concentrations of Copper and Zinc Salt Solutions:

The construction of the Daniell cell using the following concentrations of Copper and Zinc salt solutions will be performed and the voltage of the cells will be entered into the Table 3.

Calculation of E_{cell} by Nernst Equation

For example, if $[Zn^{2+}]$ and $[Cu^{2+}]$ are given as 0.05 M and 0.01 M, respectively, then

$$E_{Cell} = E_{Cell}^{o} - \frac{RT}{nF} \ln \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$R = 8.314\ J \cdot K^{-1} \cdot mol^{-1}$;
$T = 303\ K$;
$n = 2$;
$F = 96500$ (constant)

$$= 1.1 - \frac{8.314 \times 303}{2 \times 96500} \ln \frac{[0.05]}{[0.01]}$$

$$= 1.1 - 0.01305 \ln \frac{[0.05]}{[0.01]}$$

$$= 1.1 - (0.01305 \times 1.6094)$$

$$= 1.1 - 0.021$$

$$E_{Cell} = 1.079\ V \text{ at } T = 303\ K$$

In a similar way, **E_{cell} values** for other concentrations are to be calculated at specific temperatures.

Calculation of ΔG_1 (at T_1 = 30 °C) and ΔG_{1a} (at T_{1a} = 50 °C) based on the E_{Cell} measurements

$\Delta G_1 = - n F E_{Cell}$ $n = 2;\ E = 0.99\ V;\ F = 96500$

$= - 2 \times 96500 \times 0.99$

$= - 191$ KJ/ mol at $T_1 = 30^{o}C$

$\Delta G_{1a} = - n F E_{Cell}$ $n = 2;\ E = 1.020\ V;\ F = 96500$

$= - 2 \times 96500 \times 1.020$

$= - 197$ KJ/ mol at $T_{1a} = 50^{o}C$

In a similar way, **ΔG_2, ΔG_{2a}, ΔG_3, and ΔG_{3a} are to be calculated from EMFs for other metal ion cocnetrations.**

Table 3: EMF of Daniell Cell observed (by calculation and experiment) from two different concentrations of zinc and copper solutions

$[Cu^{2+}]$	$[Zn^{2+}]$	E_{cell} (calculated by Nernst Eqn)	E_{cell} (experimental)	% Error	Free-energy change (ΔG) or W_{max} (KJ/ mol)
At Room Temperature, T_1 = 30 ºC/ 303 K					
0.01 M	0.05 M				
0.05 M	0.01 M				
0.1 M	0.1 M				
At T_{1a} = 50 ºC/ 323 K					
0.01 M	0.05 M				
0.05 M	0.01 M				
0.1 M	0.1 M				

Part-C: Calculation of ΔH and ΔS at 40 ºC / 313 K based on the T_1 and T_{1a} parameters

ΔG_1 at 30 ºC = -191 KJ/ mol and ΔG_{1a} at 50 ºC = -197 KJ/ mol, Then

$$\Delta G^{1*} \text{ at } 40\ ^{o}C\ (T = 313\ K) = \frac{-191-197}{2} = -194\ KJ/\ mol$$

ΔG_2 at 30 ºC = -195 KJ/ mol and ΔG_{2a} at 50 ºC = -199 KJ/ mol, Then

$$\Delta G^{2*} \text{ at } 40\ ^{o}C\ (T = 313\ K) = \frac{-195-199}{2} = -197\ KJ/\ mol$$

ΔG_3 at 30 ºC = -206 KJ/ mol and ΔG_{3a} at 50 ºC = -208 KJ/ mol, Then

$$\Delta G^{3*} \text{ at } 40\ ^{o}C\ (T = 313\ K) = \frac{-206-208}{2} = -207\ KJ/\ mol$$

According to the Gibbs-Helmholtz equation, the ΔH can be calculated for Daniell Cell having $[Zn^{2+}]/[Cu^{2+}]$ = [0.05 M]/ [0.01 M],

$\Delta G = \Delta H + T\ [(\partial(\Delta G))/\partial T]_P$, then

$(\partial(\Delta G))/\partial T = (\Delta G_{3a} - \Delta G_3)/(T_2 - T_1)$

$= (-197-(191))/(323-303) = (-6)/20 = -0.3$ KJ/ K

By substituting $(\partial(\Delta G))/\partial T$ in Gibbs – Helmholtz eqn

We can get,

$\Delta H = \Delta G - T\,[(\partial(\Delta G))/\partial T]_P$

$= -194 - [(313) \times (-0.3)]$

$= -194 + 93.9$

$= -100.1$ KJ

Calculation of ΔS from ΔG and ΔH

$\Delta G = \Delta H - T\,\Delta S$

$\Delta S = (\Delta H - \Delta G)/T$

$= (-100.1 - (-194))/313$

$\therefore \Delta S = 93.9/313 = 0.3$ KJ.K^{-1}.mol^{-1}

In a similar way, the **ΔH and ΔS for other concentrations of [Zn^{2+}] and [Cu^{2+}] as given in Table-4 are to be calculated.**

Table-4: Final Results: At 313 K (40 ºC)

[Cu^{2+}]	[Zn^{2+}]	ΔG, KJ/ mol	ΔH, KJ/ mol	ΔS, KJ·K^{-1}·mol^{-1}	Marks Awarded
0.01 M	0.05 M				
0.05 M	0.01 M				
0.1 M	0.1 M				

Experiment 5. Determination of reaction rate, order and molecularity of ethylacetate hydrolysis

Introduction:

Esters are chemical compounds that are derived from organic or inorganic acids were at least one hydroxyl group is replaced by an alkoxy group. Esters with low molecular weight are found in pheromones and essential oils. They are less polar than alcohol and more polar than ethers. They are characterized by some water solubility due to their participation in hydrogen bonding. In easy terms, Hydrolysis is a simple reaction with water.

Acid-Catalysed Hydrolysis of Methyl Esters

Since water is such a poor nucleophile, the method of increasing the rate of nucleophilic addition with an ester is to increase the electrophilicity of the ester. This can be achieved in a number of ways. The most common is to use a Lewis acid or Bronsted acid to form a positively charged intermediate that is far more reactive and even mild nucleophiles such as water will react.

Importance:

- Sodium acetic acid derivation is a kind of salt. At the point when water is added to sodium acetic acid derivation, the compound securities separate, making it isolate into sodium particles and acetic acid secondary particles. At that point, acetic acid secondary particles in the water then join with hydrogen iotas to make a corrosive called acidic corrosive.
- Alkyl halides are synthetic mixes frequently utilized as a part of refrigerants as CFCs (chlorofluorocarbons). At the point when water is added to alkyl halides, the substance is then changed over to a kind of liquor, which is for the most part more secure for the earth.
- Through the study of chemical kinetics, it can reasonably guide us to know how to control the reaction conditions and improve the main reaction rate, in order to increase the production of chemical products, and also guide us to learn how to suppress or slow down the reaction rate of side reactions

Expt. No.: **Date:**

Experiment	**Determination of Reaction Rate, Order and Molecularity of Ethyl acetate Hydrolysis**
Problem definition	Acid catalyzed hydrolysis of ethyl acetate to produce acetic acid and ethyl alcohol
Methodology	Following chemical kinetics of the reaction through acid base titrimetric
Solution	Hydrolysing ethyl acetate to get ethyl alcohol and acetic acid
Student learning outcomes	Follow the hydrolysis through pseudo first order kinetics to know the reaction rate, order and molecularity

Aim:

To determine the rate constant of the hydrolysis of Ethyl acetate using an acid as catalyst

and to understand the order and molecularity of the reaction.

Principle:

The hydrolysis of an ester occurs according to the equation

$$CH_3COOC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$$

This reaction follows pseudo first order kinetics since the concentration of ethyl acetate only changes with time and the concentration of water does not change significantly since it is taken in large excess. **Two molecules** are involved in the reaction but the order is first only and the rate is computed using **first order rate equation**.

The following rate equation is applicable to the above reaction:

Rate α [Ester] [H_2O]

Since [H_2O] remains constant,

Rate = $\frac{-dc}{dt} = k'_1\,[Ester]$

where 'c' represents concentration of the ester at any time t; k'_1 is the specific velocity constant. As the reaction progresses, the accumulation of acetic acid increases. Drawing a known volume of the reaction mixture at known regular intervals of time and titrating it against standard sodium hydroxide solution.

The acid hydrolysis of ester is a first-order reaction, bimolecular reaction and the reaction proceeds pseudo first order kinetics. When the amount of water is relatively large, the reaction goes practically to completion (the equilibrium shifts to the right) and the rate is first order with respect to the ester. The hydrolysis takes place slowly with pure water and is catalyzed by acids.

Requirements:

Reagents and solutions: Ethyl acetate, 0.5N HCl, 0.2N NaOH, Phenolphthalein indicator, Ice-cubes

Apparatus: Burette 50mL, Pipettes-5mL, 10mL, Conical flasks, Wash bottle Reaction bottle 250mL.

Procedure:

Exactly 100 mL of 0.5N-hydrochloric acid solution are taken in a 250 mL clean reaction bottle and exactly 5 mL of the ester is added to it. Zero time is noted when half the volume of ester solution in the pipette is transferred into the reaction bottle. After thorough mixing, immediately 10mL of the solution is pipetted out into a clean conical flask containing ice-cold water. It is then titrated against 0.2 N sodium hydroxide solution from the burette using Phenolphthalein indicator. The end-point is the first appearance of a pale permanent

pink colour. The same volumes of the reaction mixture is withdrawn at regular intervals, say 10 minutes and is titrated against sodium hydroxide solution. The reaction is allowed to go to completion by keeping the reaction Mixture over a hot water bath for about 90 minutes. The final reading is then noted.

Calculations:

Let V_0 be the volume of alkali used at zero time and V_t be the volume of alkali used after the time 't' seconds. Let V_∞ be the reading when the reaction is completed. Concentrations of the ester at various time intervals are expressed in terms of volume of NaOH solution.

A = initial concentration of ester = $(V_\alpha - V_o)$

(a-x) = concentration of ester at any time't' = $(V_\alpha - V_O) – (V_t – V_o)$

= $(V_\alpha - V_o – V_t + Vo)$

= $(V_\alpha - V_t)$

The specific rate constant of the reactions is given by

$$k_1' = \frac{2.303}{t} \log\left(\frac{a}{a-x}\right)$$

$$k_1' = \frac{2.303}{t} \log\left(\frac{V_\alpha - V_o}{V_\alpha - V_t}\right)$$

k_1' = Slope x 2.303

The rate constant values are calculated at different time intervals which should nearly be the same. A graph is drawn between **$\log(V_\alpha - V_t)$** and the Time'**t**'. From the slope of the plot, the rate constant is calculated and it is compared with the experimental value.

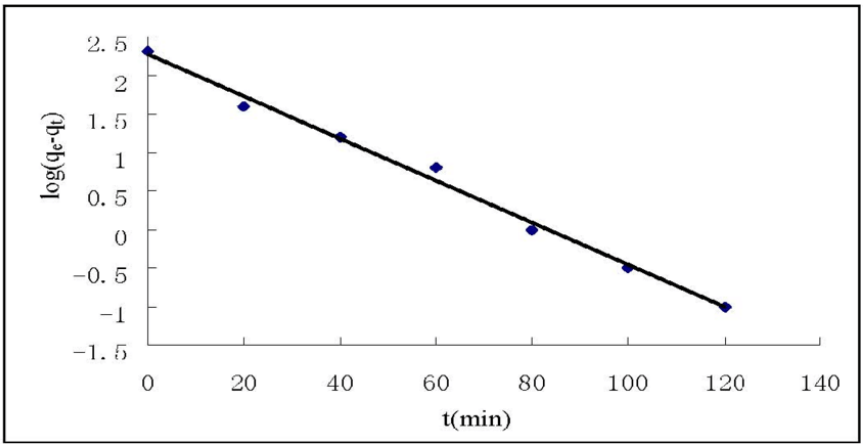


Table-1:

S.No.	Time min	Volume of NaOH mL	$(V_\infty - V_t)$ mL	$log\ (V_\infty - V_t)$	$K = \frac{2.303}{t} \log \frac{(V_\infty - V_0)}{(V_\infty - V_t)}$ min^{-1}
1	0				
2	10				
3	20				
4	30				
5	40				
6	50				
7	60				
8	∞				

Results:

The Rate Constant for the hydrolysis of an ester from

1. Calculated value =
2. Graphical value =
3. Molecularity of the reaction =
4. Order of the reaction =

Experiment 6. Magnesium Salt Water Activated Battery- Effect of Salt Concentration on Voltage generation

1. **Introduction:**

Magnesium batteries are batteries that utilize magnesium cations as the active charge transporting agent in solution and as the elemental anode of an electrochemical cell. Both non-rechargeable primary cell and rechargeable secondary cell chemistries have been investigated. A significant advantage of magnesium cells is the use of a solid magnesium anode, allowing a higher energy density cell design.

The voltage created in batteries is due to movement of electron from anode to cathode. At the anode oxidation takes place releasing electrons and these electrons travel to the cathode and at the cathode reduction takes place. The electrical energy is generated by the movement of electrons. When a metal electrode such as magnesium is immersed in the electrolyte containing NaCl, it will lose two electrons and act as anode and oxidation will take place, i.e., Mg will be converted to Mg^{+2} and the electrons will pass into the electrolyte. The electrons travel through the electrolyte towards copper electrode making it more negative and hence reduction will take place at cathode (i.e., at the copper electrode). Since these electrodes are of different metals, their reactivities will be different, and so different. One electrode will become charged to a greater extent than the other and because of the difference in reactivities will have a different voltage. Because of this difference, electrons want to move from one electrode to the other. This attractive potential is the voltage that we measure between the electrodes, and the origin of the batteries' electrical power.

Experiment	**Construction of Mg – salt water activated battery Effect of Salt Concentration on Voltage generation**
Problem definition	Construction of Mg – Salt water activated battery using Copper as cathode, measure its voltage and verify performance.
Methodology	Construction of a galvanic cell using Mg as anode and Copper as as cathode and measure its voltage as individual cell and when cells are connected in series using salt water as electrolyte.
Solution	To verify working of Mg-Cu salt water activated battery and its use as an eco-friendly galvanic cell.
Student learning outcomes	Understand the construction of Mg anode and Copper cathode as to make Mg-Cu salt water activated primary battery and understand its utility for low current applications.

2. **Aim:**

   To construct galvanic cells using Mg as anode and Cu as cathode and salt water as electrolyte and to understand its utility for low current applications.

3. **Working principle of the system:**

When a highly oxidizable electrode such as Magnesium is used as anode, it loses two electrons easily to the electrolyte i.e. NaCl (salt water) and gets oxidized to Mg^{2+}. These electrons travel through the electrolyte and get discharged at the cathode i.e. Copper where Cu^{2+} gets reduced to metallic copper.

- **At anode**

$$Mg \rightarrow Mg^{+2} + 2e^{-} \quad \text{(OXIDATION)}$$

- **At Cathode**

$$Cu^{2+} + 2e^{-} \rightarrow Cu \quad \text{(REDUCTION)}$$

Movement of two electrons from anode to cathode results in production of electrical energy.

Requirements:

Materials:

Magnesium anodes (6 nos.) Copper wire cathodes (6 nos.), plastic containers of 100 mL capacity (6 nos.), Crystal salt (30 grams), Alligator clips, Copper wire, Digital multimeter (DMM) and 6V LED bulb.

4. **Methodology :**

Experimental Setup:

- Six empty plastic cups of 100 mL capacity are taken.
- 100 ml of NaCl solution containing 5% of crystal salt is filled in each one of the cups. This is made by dissolving 5 grams of crystal salt in 100 mL of water (30 grams of salt in 300 mL of tap water)
- One Mg anode connected with copper leads through a glass tube and one Copper Cathode connected with copper leads through a glass tube are inserted into the pre-made holes on the lid. The distance between Mg and Cu is maintained at 2 cms.
- The lid with the electrode assembly is fixed by threading to the plastic cup. All six cups are arranged in similar manner.
- The Mg and Cu electrode should dip in salt water solution completely.
- Thin copper wire connected with alligator clips is used for electrical connections. Use thin copper wire and alligator clips to connect each electrode.
- The voltage of each cell is measured by connecting Copper to positive and magnesium to negative terminal of the DMM and noted.
- Then all the six cells were connected in series by making one main Mg as anode and one Copper as cathode for the final connection.
- The voltage of the battery with six cells connected in series is measured and recorded.
- One LED bulb of 9 V rating is connected to the main terminals and this started glowing and is expected to glow for a period of 4-5 hours when all the Mg gets converted into $MgCl_2$.

5. **Observations and Results:**

Cell Voltages/Volts											
Cell 1		Cell 2		Cell 3		Cell 4		Cell 5		Cell 6	
Th.	Obs.	Th.	Obs.	Th.	Obs.	Th.	Obs.	Th.	Obs.	Th.	Obs.

%Error		%Error		%Error		%Error		%Error		%Error	

Observed Voltage of battery when all six cells are connected in series :_____V

Theoretical voltage of battery when all six cells are connected in series: _____ V

% Error :

Experiment 7. Size dependent color variation of Cu_2O nanoparticles by a spectrophotometer

Subject Code	Size dependent color variation of Cu_2O nanoparticles by a spectrophotometer	L	T	P	J	C
		0	0	2	0	1
Problem definition	Nanomaterial plays a significant role in our recent life. The electronic materials such as processors of computers, mobile phones consist of nanosized semiconductor materials whose properties depend on their composition, size. It is essential to understand the size dependent optical properties of semiconductor compound such as Cu_2O.					
Methodology	Cu_2O particles can be synthesized *via* redox reactions between sugar and Benedict's reagent. Heating the Benedict's reagent with glucose solution in alkali medium leads to reduction of blue coloured copper(II) ions to Cu_2O nanoparticle with other colour. These solution, when exposed to light, scatters light (Tyndall effect) which can be measured by Nephelometry.					
Solution	Prepare Cu_2O nanoparticle solution and measure its turbidity					
Student learning outcomes	Students will learn to a) Prepare the Cu_2O nanoparticle and its properties b) Perform Nephelometric method and size specific Tyndall effects					

Importance of the experiment:

Nanoscience and nanotechnology play an important role in our daily life. Materials from processors of personal computers, mobile phones, light emitting diodes to medical equipment, drugs etc. consist of nanosized parts with unique properties. The electronics material are mainly based on semiconductor materials such as silicon and gallium arsenide whose properties depend on their composition, shape and size. This experiment aims at providing an explanation of size-dependent optical properties of semiconductor compounds.

Optical Properties of Cu_2O:

Cu_2O absorbs light in wavelength range from UV to orange and its most common color is red. However, as Cu_2O is a semiconductor, its bandgap depends on size, shape, and crystalline structure of the sample. Below certain limit (less than micrometer), bandgap changes subsequently changing the color of Cu_2O.

Principle:

When a solution with suspension is exposed to light, part of the incident light energy is dissipated by absorption, reflection and refraction, while the remaining is transmitted. When the suspension is viewed at right angles to the direction of the incident light, the system appears opalescent due to the reflection of light from the particles of suspension (Tyndall effect). The reflection of light is irregular in nature and therefore the

term scattered light is used for it. The measurement of the intensity of scattered light as a function of the suspended particle concentration is done by the technique known as Turbidimetry or Nephelometry. The intensity of scattered light is measured at right angles to the direction of the incident light. Light scattering properties changes with respect to the particle size.

Reaction scheme:

Cu_2O particles can be synthesized *via* redox reactions between sugar and alkaline solution of copper(II) complex with citrate anions (Benedict's reagent). Heating the sugar solution with Benedict's reagent in alkali medium leads to sugar oxidation to corresponding carboxylic acid salts and reduction of blue colored copper(II) ions to Cu_2O with other color. Higher reactivity leads to formation of bigger Cu_2O particles.

Oxidation: $RCHO + 3\ OH^{\ominus} \longrightarrow RCO_2^{\ominus} + 2\ H_2O + 2e^{\ominus}$

Reduction: $2Cu^{2\oplus}_{(\text{in complex})} + 2\ OH^{\ominus} + 2e^{\ominus} \longrightarrow Cu_2O + H_2O$

Summary: $RCHO + 2Cu^{2\oplus}_{(\text{in complex})} + 5\ OH^{\ominus} \longrightarrow RCO_2^{\ominus} + Cu_2O + 3H_2O$

Reagents & Apparatus:

a) Copper sulphate ($CuSO_4 \cdot 5H_2O$)	(d) Sodium hydroxide (NaOH)
(b) Sodium carbonate (Na_2CO_3)	(e) Glucose solution
(c) Trisodium citrate dihydrate ($Na_3C_6H_5O_7 \cdot 2H_2O$) [$Na_3Cit$]	(f) Nephelometer spectrophotometer

Procedure:

(a) Preparation of Benedict's reagent: Dissolve 6.0 g of $CuSO_4 \cdot 5H_2O$ in 20 mL of hot water. Prepare alkaline citric solution by mixing 4.0 g of Na_2CO_3 and 7.0 g of Na_3Cit in 40 mL of water. Add this citrate solution in hot $CuSO_4$ solution. After that the mixture was diluted with 40 mL of water. [Conc. of stock solution = 240 mmol w.r.t. $CuSO_4$.]

(b) Preparation of glucose solution: Prepare glucose solution by dissolving 30.0 g glucose in 750 mL of water. [Conc. of stock solution = 222 mmol]

(c) Preparation of NaOH solution: Prepare 100 mL of 1.0 M NaOH solution.

(d) Preparation of conditioning agent (Glycerol-alcohol solution): Mix 400 mL absolute ethanol to 200mL of glycerol.

(e) Preparation of Cu_2O solution:

1. Take 50 mL volumetric flask and label them **1, 2, 3, 4, 5 (for Table 1)** & **A, B, C, D (for Table 2)**

Table 1:

Sample No.	Vol. of Benedict's reagent (mL)	Vol. of glucose solution (mL)	Conc. of NaOH (M)	Observation/Color
1	0.5	4.5	0.0	Yellow

2	0.5	4.5	0.0001	
3	0.5	4.5	0.001	
4	0.5	4.5	0.01	
5	0.5	4.5	0.1	Red

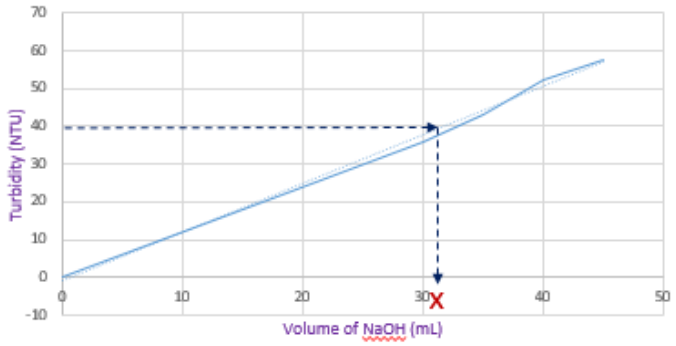
Table 2:

Sample No.	Vol. of Benedict's reagent (mL)	Vol. of glucose solution (mL)	Vol. of NaOH **0.01 M** (mL)
A	0.5	4.5	45
B	1.0	9.0	40
C	1.5	13.5	35
D	2.0	18.0	30

2. Mix certain amounts of Benedict's reagent in the volumetric flask and dilute the solution with 45 mL of NaOH solution of certain concentration.
3. To each flask add glucose solution and make the volume to 50 mL (with respective NaOH solution) and mix well.
4. From the flask, take around 10 mL solution in test tubes and place the test tubes with reaction mixtures into a beaker with boiling water and keep it there for 5 min.
5. After that, transfer the test tubes to the beaker with room temperature water in order to stop reaction.
6. The fresh samples can be photographed (Table 1) and use for analysis purpose (Table 2).
7. Next, mix equal volume of conditioning agent (glycerol:ethanol – 1:2) to the sample solution (10mL of analyte + 10 mL of conditioning agent). Afterwards, transfer the samples (**A – D**) in cuvette (the sample tube for turbidity analysis that should be very clean inside and outside and should not contain any scratches). Note the reading on turbidity scale [Turbidity is measured in NTU (Nephelometric Turbidity Units)].
8. Prepare a standard curve by plotting turbidity (NTU units) vs the volume of NaOH (Refer the Appendix in presentation file, .ppt). Then calculate the unknown concentration of copper(II) by back calculation.

Observations:

Sample	**Vol. of NaOH (mL)**	**Turbidity (NTU)**
A		
B		
C		
D		



Result:

1) The volume of NaOH for the unknown solution _____ mL.

2) The concentration Cu(II) solution is _____ mmol.

Reference:

N. E. Markina, M. V. Pozharov, A. V. Markin. J. Chem. Educ. **2016**, 93, 4, 704–707.

Experiment 8. Synthesis of Al-doped ZnO Nanoparticles via Sol-Gel Method

Importance of Al-Doped ZnO Nanoparticles for Optoelectronic Applications

Zinc oxide (ZnO) is a versatile II-VI semiconductor with a wurtzite structure, known for its wide bandgap (3.37 eV), high exciton binding energy (60 meV), and excellent optical transparency. These properties make it ideal for optoelectronic devices, including solar cells, LEDs, sensors, and piezoelectric applications. However, pure ZnO suffers from high electrical resistivity, which limits its efficiency in conductive applications. Doping ZnO with aluminum (Al) enhances its electrical conductivity while maintaining optical transparency, making Al-doped ZnO (AZO) a superior alternative to costly and toxic indium tin oxide (ITO). Al^{3+} ions substitute Zn^{2+} sites, introducing free electrons that reduce resistivity and modify the band structure, enabling tunable optoelectronic properties.

Additionally, AZO exhibits improved light trapping, plasmon resonance, and photoluminescence, beneficial for transparent electrodes, gas sensors, and scintillation devices. A major challenge in nanoparticle synthesis is agglomeration, which reduces surface reactivity. Using polyvinyl alcohol (PVA) as a capping agent helps control nucleation, yielding uniform, stable nanoparticles with enhanced optical performance. Recent studies show that PVA-capped ZnO exhibits unique wavelength-dependent photoluminescence, further expanding its applications in bioimaging and nanophotonics. This study focuses on optimizing Al doping in ZnO to improve structural and optical properties while addressing discrepancies in reported bandgap shifts. By systematically analyzing dopant effects, we aim to develop high-performance AZO nanomaterials for next-generation optoelectronic and energy-harvesting technologies.

Key Highlights:

Wide bandgap & high exciton stability make ZnO ideal for optoelectronics.

Al doping reduces resistivity while maintaining transparency.

AZO is a cost-effective, non-toxic alternative to ITO.

Objective

To synthesize pure and Mg-doped ZnO ($Mg_xZn_{1x}O$) nanoparticles using the sol-gel method and characterize their structural and optical properties.

Materials Required

Chemicals:

1. Zinc nitrate hexahydrate ($Zn(NO_3)_2 \cdot 6H_2O$) – AR Grade
2. Magnesium nitrate hexahydrate ($Al(NO_3)_3 \cdot 6H_2O$) – AR Grade
3. Polyvinyl alcohol (PVA)
4. Deionized water

Equipment:

1. Magnetic stirrer with hot plate
2. Weighing balance
3. Beakers (100 mL, 250 mL)
4. Measuring cylinder (50 mL)

5. Glass rod

6. Oven (for drying)

7. Muffle furnace (for annealing)

8. Crucible

Procedure

Step 1: Preparation of Precursor Solutions

1. Metal Nitrate Solution Preparation:

- Weigh $Zn(NO_3)_2 \cdot 6H_2O$ and $Al(NO_3)_3 \cdot 6H_2O$ in stoichiometric ratios to obtain 1 g of final product ($Al_xZn_{1x}O$, where x = 0.00, 0.01, 0.02, 0.03, 0.04, 0.05).

- Dissolve the weighed nitrates in 15 mL of deionized water under continuous stirring (~30 min) to obtain a clear solution.

2. PVA Solution Preparation:

- Weigh 3 g PVA and dissolve it in 30 mL deionized water under continuous stirring (~3–4 h) at 70°C until a homogeneous milky-white solution is obtained.

Step 2: Sol-Gel Formation

1. Slowly add the metal nitrate solution dropwise into the PVA solution at 70°C under constant stirring.

2. Continue stirring until a viscous gel is formed (~1–2 h).

Step 3: Drying and Annealing

1. Transfer the gel into a crucible and dry it in an oven at 150°C for 12 h to remove moisture and obtain a dried powder.

2. Anneal the dried powder in a muffle furnace at 600°C for 2 h with a heating rate of 2°C/min.

3. Allow the furnace to cool naturally to room temperature.

Step 4: Sample Labeling

Label the annealed samples as follows:

- Undoped ZnO (x = 0.00)

- Al1 (x = 0.01)

- Al2 (x = 0.02)

- Al3 (x = 0.03)

- Al4 (x = 0.04)

- Al5 (x = 0.05)

- Al6 (x = 0.06)

Precautions

1. Use gloves and lab coats while handling chemicals.

2. Ensure proper ventilation when heating PVA to avoid fume inhalation.

3. Maintain constant stirring to avoid precipitation.

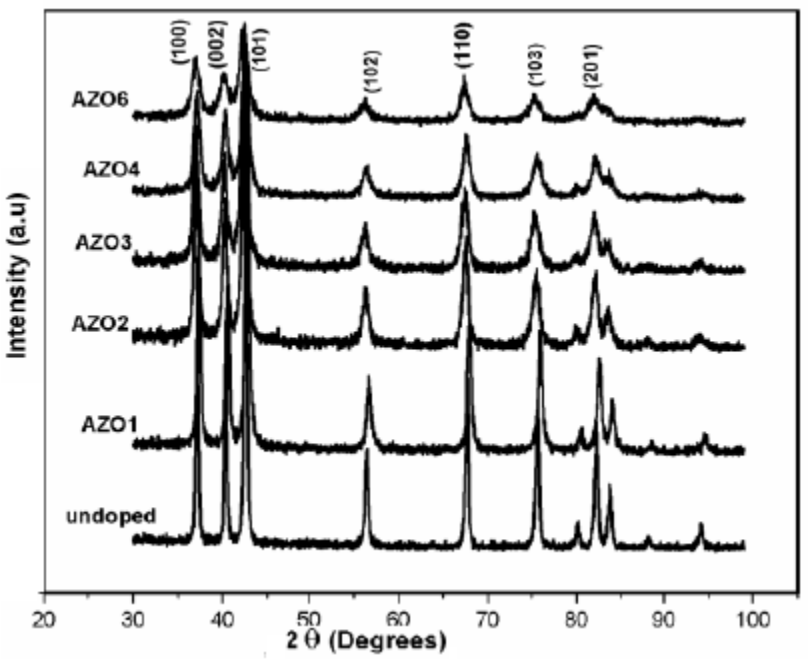
4. Control the heating rate during annealing to prevent thermal shock.

Expected Outcomes

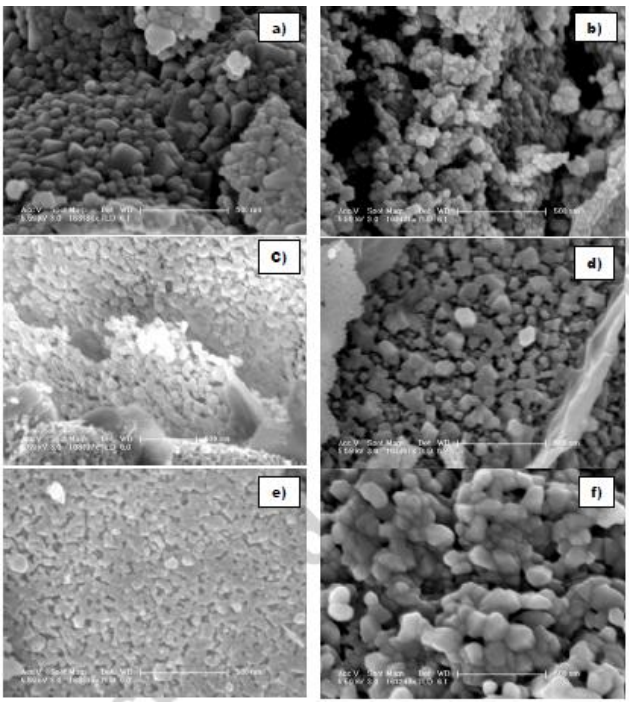
- Formation of white/off-white powder (Al-doped ZnO nanoparticles).

- The crystallinity and doping efficiency can be confirmed via XRD, SEM, and UV-Vis spectroscopy.

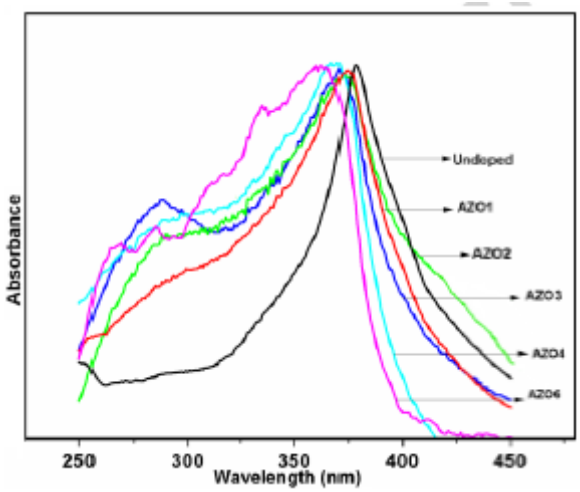
XRD:



SEM:



UV-Vis spectroscopy:



References:

1. A.N. Mallika, A. Ramachandra Reddy, K. Sowri Babu, K. Venugopal Reddy, Synthesis and optical characterization of aluminum doped ZnO nanoparticles, Ceramics International, Volume 40, Issue 8, Part A, 2014, Pages 12171-12177, ISSN 0272-8842,

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Experiment 9. Kinetic Study of Rhodamine B Degradation by •OH Radicals (H_2O_2/UV System)

Importance of Studying Dye Degradation via UV/H_2O_2 Advanced Oxidation

This experiment investigates the degradation of synthetic dyes (e.g., Rhodamine B) using UV-activated hydrogen peroxide (UV/H_2O_2), a critical Advanced Oxidation Process (AOP) for wastewater treatment. Industrial effluents, especially from textile operations, contain persistent and toxic dyes that resist conventional methods like coagulation or biological treatment. These pollutants endanger aquatic ecosystems and human health due to their mutagenic potential. The UV/H_2O_2 system offers a sustainable alternative by generating hydroxyl radicals (•OH)—powerful oxidants (2.8 eV) that break down complex dyes into harmless compounds like CO_2 and water, without secondary sludge.

Understanding this process is essential because:

1. Environmental Relevance: Azo dyes, widely used in textiles, often survive traditional treatments. UV/H_2O_2 achieves rapid decolorization and partial mineralization, reducing environmental impact.

2. Mechanistic Insight: The experiment demonstrates pseudo-first-order kinetics, highlighting how •OH radicals attack dye molecules, enabling optimization of degradation rates.

3. Practical Advantages: Unlike Fenton reactions (pH-dependent) or photocatalysis (costly catalysts), UV/H_2O_2 operates at neutral pH and requires no catalyst recovery, simplifying scale-up.

4. Skill Development: Trains students in kinetic analysis, spectrophotometry, and AOP design—key competencies for environmental engineering.

By varying H_2O_2/dye concentrations and monitoring absorbance decay, this lab bridges theory and real-world applications, emphasizing how AOPs can mitigate industrial pollution. The results will underscore the balance between reaction efficiency and operational constraints, preparing students to innovate in sustainable wastewater management.

Key Learning Outcomes:

a. Role of •OH radicals in pollutant degradation.

b. Kinetics of pseudo-first-order reactions.

c. Trade-offs in AOP selection for industrial wastewater.

Handle H_2O_2 with care and wear UV-protective gear. Record all data for comparative analysis.

Why This Matters:

This lab replicates challenges faced by treatment plants, equipping future scientists to combat water pollution with cutting-edge oxidative techniques.

Objective

To investigate the pseudo-first-order kinetics of Rhodamine B (RhB) degradation via hydroxyl radical (•OH) attack generated from UV-activated H_2O_2.

Required Materials:

Chemicals

1. Rhodamine B (RhB), 2. Hydrogen Peroxide (H_2O_2, 35% w/w), 3. Deionized Water, 4. Quartz cuvettes (1 cm path length)

Equipment

1. UV-Vis Spectrophotometer, 2. UV Lamp (UVGL-58, 6W, 254 nm), 3. Magnetic Stirrer & Beakers (50–250 mL), 4. Pipettes, Volumetric Flasks, 5. Thermometer (for 25±2°C control)

Procedure

Step 1: Preparation of Solutions:

1. RhB Stock Solution (1×10^{-3} M):
   - Dissolve 0.0479 g RhB in 100 mL deionized water (250 mL flask).
2. H_2O_2 Working Solution (1.67 mM):
   - Dilute 0.16 mL of 35% H_2O_2 in 100 mL water (prepare fresh).

Step 2: Reaction Setup

1. Control Experiments:

   Test RhB degradation under:
   - UV alone (no H_2O_2).
   - H_2O_2 alone (no UV).
2. H_2O_2/UV Degradation:
   - Mix 50 mL RhB solution + 1.67 mM H_2O_2 in a beaker.
   - Place 4 cm from UV lamp (254 nm mode).
   - Stir continuously at 25±2°C.

Step 3: Kinetic Monitoring

1. At 5-min intervals, withdraw 3 mL aliquot.
2. Measure absorbance at 543 nm (UV-Vis).

3. Plot ln(A_t/A_0) vs. time to determine rate constant (k).

Reaction Mechanism:

1. •OH Generation:

$$H_2O_2 + h\nu \rightarrow 2\dot{O}H$$

2. Dye Degradation:

$$\dot{O}H + RhB \rightarrow \dot{P} \text{ (Intermediates)} \rightarrow \text{Colorless Products}$$

Data analysis

1. Pseudo-First-Order Kinetics:

$$ln\left(\frac{A_t}{A_0}\right) = -kt$$

$$\text{-slope} = k(\text{rate constant})$$

2. % of Decoloration:

$$\%\ \text{Decoloration} = \left(1 - \frac{A_t}{A_0}\right) \times 100$$

Expected Results

1. The linear ln(A_t/A_0) vs. time plot confirms pseudo-first-order kinetics.
2. Rate constant (k) increases with [H_2O_2], decreases with [RhB].

Safety & Precautions

1. Wear gloves/goggles – H_2O_2 is corrosive.
2. UV shielding – Avoid direct exposure.
3. Dispose waste in labeled hazardous containers.

References

1. Muruganandham & Swaminathan (2006), J. Hazard. Mater.
2. Xu & Wang (2012), *J. Environ. Sci.

Appendix

1. Table 1: k values at varying [H_2O_2].
2. Fig. 1: Absorbance decay curves.

Table 1: Effect of H_2O_2 Concentration on RhB Degradation Kinetics

(Fixed [RhB] = _____ M, Temp = 25±2°C, λ = 543 nm)*

[H_2O_2] (mM)	**Time (min)**	**Absorbance (A_t)**	**ln(A_t/A_0)**	**% Decoloration**	**Notes**
0	0				Control (No H_2O_2)
0	5				
0	10				
1.67	0				
1.67	5				
1.67	10				

Calculations:

ln(A_t/A_0) = ln(Current Absorbance / Initial Absorbance).

% Decoloration $= \left(1 - \frac{A_t}{A_0}\right) \times 100$

Table 2: Effect of RhB Concentration on Degradation Kinetics

(Fixed [H_2O_2] = 1.67 mM, Temp = 25±2°C, λ = 543 nm)

[RhB] (M)	**Time (min)**	**Absorbance (A_t)**	**ln(A_t/A_0)**	**% Decoloration**	**k (min^{-1})**
1.00E-04	0				
1.00E-04	5				
1.00E-04	10				
1.00E-03	0				
1.00E-03	5				
1.00E-03	10				

Note:

k = Slope of ln(A_t/A_0) vs. time plot (from linear regression).

Table 3: Summary of Rate Constants (k) and Decoloration Efficiency

Condition	$[H_2O_2]$ (mM)	[RhB] (M)	k (min^{-1})	% Decoloration (30 min)
UV only	0	1.00E-04		
H_2O_2 only	1.67	1.00E-04		
H_2O_2 + UV (Low RhB)	1.67	1.00E-04		
H_2O_2 + UV (High RhB)	1.67	1.00E-03		

Table 4: Calibration Curve for RhB

(For standard concentration-absorbance relationship)

RhB] (M)	Absorbance (543 nm)
0	
5.00E-05	
1.00E-04	
5.00E-04	
1.00E-03	

Plot: Absorbance vs. [RhB] to verify linearity (Beer-Lambert law).

Notes for Tables:

1. Record triplicate measurements for accuracy.
2. Label samples (e.g., "Run 1, Run 2").
3. Include errors (±) if replicates are performed.

References:

1. Fatima H. AlHamedi, M.A. Rauf, S. Salman Ashraf, “Degradation studies of Rhodamine B in the presence of UV/H_2O_2 ,Desalination, Volume 239, Issues 1–3, 2009, Pages 159-166, ISSN 0011-9164, DOI: https://doi.org/10.1016/j.desal.2008.03.016
2. Abbas Mehrdad, Bakhshali Massoumi, Robab Hashemzadeh, Kinetic study of degradation of Rhodamine B in the presence of hydrogen peroxide and some metal oxide, Chemical Engineering Journal, Volume 168, Issue 3, 2011, Pages 1073-1078, ISSN 1385-8947, DOI: https://doi.org/10.1016/j.cej.2011.01.087.

Experiment 10. Determination of State of Charge (SoC)

Importance of SoC:

1. As we know, unlike the Li-ion batteries in other countries, in India, the batteries often experience issues such as explosion due to poor thermal management and battery assembly during usage. Hence, it is very important to understand the battery chemistry before assembling the bulk battery pack.
2. The existing battery management and assembly system is not suitable for all conditions due to the poor understanding of battery chemistry.
3. The state of charge is the vital one for all kinds of battery manufacturing. Hence, it is important to understand battery chemistry because the state of charge is really important.
4. Currently, the successful battery cathode material or battery cathodes available in the market are Lithium iron phosphate and Lithium Nickel Manganese Oxide-based cathodes. However, there are still lots of other types of battery chemistry (cathode materials available). Hence, it is very important to understand the battery chemistry of Li-ion batteries.

Aim:

The evaluation of the state of charge (SoC) for the Li-ion battery using different types of battery cathode materials (various battery chemistries) through the open circuit voltage (OCV), current, and resistance with respect to the discharging and rest time.

Principle:

State-of-charge is generally defined as the available amount of charge in a given battery (*Q*) related to the maximum available amount of charge, which can be taken from this battery after a 100% full charging (*C*), and is usually expressed as a percentage:

$$SoC = \frac{\text{available amount of charge (Q)}}{\text{Maximum available amount of charge (C)}} \times 100$$

The amount of charge is calculated by the capacity of a battery (C).

C = current x time (holding time for the given current)

The available amount of charge is the current available at the given time (Current x Time)

Apparatus required:

1. The equipment and connectors required for this experiment
2. The batteries of Lithium iron phosphate and lithium manganese oxide cathodes-based battery of either 18650 or 23650 model.
3. The holder is to get the positive and negative connections.



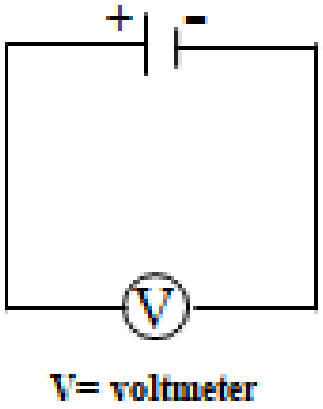
4. The light-emitting diode of a 2.5 to 3.7-volt range of operations

5. A voltmeter and an ammeter  

6. A button  switch

Procedure:

Assemble the single cell battery of two different chemistries, say A and B, with the respective holders, switches, and LED load. To measure the open circuit voltage, we will connect the batteries in series to the multimeter (see Fig.1) and measure the voltage initially and take it over time at o. Then arrange the multimeter in such a way to measure the flow of electrons and switch on the biased LED load as shown in the figure, and measure the current. After 5 minutes, switch off the circuit and rest for 5 minutes. Then measure the OCV at time 5 minutes and subsequently proceed to take the reading for 30-45 minutes and tabulate the results with OCV, current, and SoC for the taken time in **Table 1**.



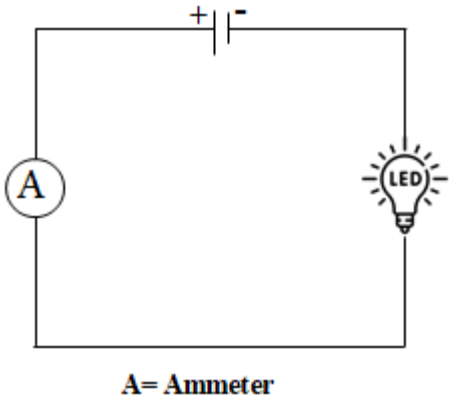


Fig.1 Circuit diagram to measure Voltage

Fig. 2 Circuit diagram to measure the current

Table: Estimation of SoC using the measured OCV and Current

Time (min)	OCV (V)		Current (mA)		Resistance(Ω)		Rest time		SoC (%) = current x time x 100 / Rated capacity	
	A	B	A	B	A	B	A	B	A	B

Results:

1.The type of chemistry in the battery A and B

2. The SoC vs OCV curve explain the amount of Li-intercalation

